NUMERICAL SOLUTION OF EUROPEAN PUT OPTION WITH CONTINUOUS DIVIDENDS AND STOCHASTIC VOLATILITY

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Abstract. In this paper, the effect of strike price, interest rate, dividends and maturities on European put option is discussed, forecasting the volatility for the data of ONGC Ltd. listed in National Stock Exchange, India, during 03-01-2000 to 30-03-2009 by EGARCH method. The option price and Greeks are determined by solving modified Black-Scholes partial differential equation by adjusting forecasted volatility at each grid point of finite difference method. It is observed that the option price increases as strike price and time of maturity increase, whereas the effect of interest rate and dividends on option price seems insignificant but play central role in portfolio management.

Introduction

The seminal Black and Scholes research work calibrated to price for options on a non-dividend paying stocks, was extended and generalized by Marton[14] for pricing of European options on continuous dividends paying stocks assuming that most common stocks pay dividends. Almost stocks pay discrete dividends rather than continuous dividends, but valuation of put options on stocks which pay discrete dividends is a intriguing problem. Inspite of that a lot of attention have been paid by researcher and financial analysts [see Roll[15], Blomeyer[16], Barone[1], Brodei[2]-[3], Wensheng and Zhen[19], Wensheng et.al.[18], Chance et.al.[4] Chung and Shackleton[5], Kallast and Kivinukk[13]and Company et. al.[6]].

Recently, Company et.al.[6], Ballester et.al.[11] used Dirac delta function, Mellin transform and semidiscretizetion technique for the pricing of option with discrete dividends. Dai T.S.[8] and Dia and Lyuu[9] introduced stair tree approach and setup the analytic formula which converts the discrete dividends model to continuous dividend model representing a function of discrete dividends and stock returns.

As a matter of fact that volatility is the main ingredient in the option pricing. But in these studies volatility is regarded as constant or a fixed in a range which is unrealistic. Recently, Rana and Ahmad[17] introduced an alternative approach for the option pricing using forecasted volatility instead of constant volatility. In this paper, volatility forecasted by EGARCH(1,1) model and governing partial differential equation is solved by Crank-Nicolson finite difference method.

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1. Formulation and Solution of Model

Suppose S and D be asset price and constant continuous dividend yield within interval dt respectively. Then Geometric Brownian Motion model for asset price is

$$\frac{dS}{S} = (\mu - D)dt + \sigma dx \tag{1.0.1}$$

where μ and σ^2 are expected rate of return and variance rate on the price ratio respectively and dx is standard Wiener process.

The modified Black-Scholes partial differential equation for European put option is

$$\frac{\partial V}{\partial t} + (r - D)S\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0, \qquad [0, \infty] \times [0, T] \tag{1.0.2}$$

where, $\sigma(S, t)$ is volatility coefficient that satisfies $\sigma(S, t) > 0$ and r is a risk-free interest rate assumed to be constant.

Suppose K and S_T be strike price and price of underlying asset on the date of expiration T respectively. The terminal payoff for the long position in European put option is defined as

$$V(S, T) = max(K - S_T, 0) \tag{1.0.3}$$

The solution domain of the problem (1.0.2) is $[0, \infty] \times [0, T]$ with final condition (1.0.3). From computational point of view, $S = \infty$ can be transform into $S = S_{max}$ in such a way that S_{max} has to be sufficiently far from region of interest in order to avoid the excessive error due to truncation and even if the imposed boundary conditions are imperfect, it does not naturally affect on the solution. Thus the computation domain of the solution of equation(1.0.2) could be $[0, S_{max}] \times [0, T]$ with right and left boundaries

$$V(0, t) = Ke^{-r(t-T)} - S_{max} \tag{1.0.4}$$

and

$$V(S_{max}, t) = 0 \tag{1.0.5}$$

The volatility (σ) forecasted by exponential GARCH (EGARCH) model developed by Nelson[10] as

$$g(\epsilon_t) = \theta\epsilon_t + \gamma[|\epsilon_t| - E(|\epsilon_t|)]$$

$$a_t = \sigma_t \epsilon_t, \qquad \ln(\sigma_t^2) = \alpha_0 + \left(\frac{1 + \beta_1 B + ... + \beta_q B^q}{1 - \alpha_1 B - .... - \alpha_p B^p}\right) g(\epsilon_{t-1})$$

where, α_0 is a constant, B is the back-shift or log operator such that $B(g(\epsilon_t)) = g(\epsilon_{t-1}$, $(1 + \beta_1 B + ... + \beta_q B^q)$ and $(1 - \alpha_1 - .... - \alpha_p B^p)$ are polynomials in B. The alternative representation for EGARCH(p,q) model is given by $a_t = \sigma_t \epsilon_t$ and

$$\ln(\sigma_t^2) = \alpha_0 + \sum_{i=1}^{p} \alpha_i \left(|\epsilon_{t-1}| + \gamma_i \epsilon_{t-1}\right) + \sum_{j=1}^{q} \beta_j \ln(\sigma_{t-j}^2)$$

where γ_i's are the leverage parameters. If $\gamma = 0$, then the good and bad news exhibit the same effect of volatility.

1.1. **Discretization of Equation.** For numerical solution of partial differential equation Eq.(1.0.1), the rectangular domain $[0, S_{max}] \times [0, T]$ is divided into $(M+1) \times (N+1)$ uniform grid points. The step widths $\triangle S$ and $\triangle t$ are in general independent, where $S_{i+1} - S_i = \triangle S = \frac{S_{max}-0}{M}$; $\quad i = 0, 1, 2, ...M \qquad t_{j+1} - t_j = \triangle t = \frac{T}{N}$; $\quad j = 0, 1, 2, ...M$ and $V_{i,j}$ denotes the numerical approximation of $V(i\triangle S, j\triangle t)$ i.e. at value of V at grid point (i, j). Keeping view for stability of finite difference scheme in mind, we used *Crank-Nicolson* method, which incorporates both explicit and implicit features([7]). The Crank-Nicolson scheme for Eq.(1.0.1) is

$$aV_j^{i-1} + bV_j^i + cV_j^{i+1} = xV_{j+1}^{i-1} + yV_{j+1}^i + zV_{j+1}^{i+1} \tag{1.1.1}$$

where

$$a = \frac{1}{4}(ri - \sigma_j^2 i^2) \triangle t \quad b = (1 + \frac{\sigma_j^2 i^2 \triangle t}{2} + \frac{r\triangle t}{2}) \quad c = -\frac{1}{4}(ri + \sigma_j^2 i^2) \triangle t$$
$$x = \frac{1}{4}(\sigma_j^2 i^2 - ri) \triangle t \quad y = (1 - \frac{\sigma_j^2 i^2 \triangle t}{2} - \frac{r\triangle t}{2}) \quad z = \frac{1}{4}(ri + \sigma_j^2 i^2) \triangle t$$

From computational point of view, the number of days for which volatility is to be forecasted are very less as compared to number of mesh points required for finite difference scheme for the same period, therefore a day is equivalent to more mesh points. Hence volatility forecasted for that particular day will be considered same for such mesh points.

1.2. **Discretization of Boundary and Final Conditions.** We need the discretization of boundary and initial conditions for the European option. For a put option, the boundary conditions are

$$V_j^0 = K \qquad and \qquad V_j^M = 0 \qquad\qquad j = 0, 1, 2, ...N$$

And Final condition

$$V_N^i = max(K - i \triangle S, 0)$$

1.3. **Stability of Scheme.** It is unconditionally stable and the amplification factor of Crank-Nicolson Scheme for Eq.(1.1.1) is

$$G(\beta) = \frac{1 - \frac{1}{2}\frac{\sigma^2 S^2}{\triangle S^2}\triangle t(1 - cos\beta) - \frac{r-D}{2}\triangle t}{1 + \frac{1}{2}\frac{\sigma^2 S^2}{\triangle S^2}\triangle t(1 - cos\beta) + \frac{r-D}{2}\triangle t}$$

which, satisfies the Von-Neumann stability condition $|G(\beta)| \leq 1$; $\quad 0 \leq \beta \leq \pi$ for any choice of $\triangle t$ and $\triangle S$, where $\beta = \lambda \triangle S$ and λ is a wavenumber. But large choice of $\triangle t$, it leads to some undesirable results. Despite this, the main appeal of the method is its second order accuracy and stability, which, are achieved with minor increase in computational cost compared with the implicit method. Crank-Nicolson scheme ultimately reduced to a sparse system of algebraic equation, whose matrix form is

$$[A](f)_j = [X](f)_{j+1} \tag{1.3.1}$$

where $[A] = [a_{mn}] \qquad [X] = [x_{mn}]$

$$[a_{mn}] = \begin{cases} a & if \quad m = n-1 \\ b & if \quad m = n \\ c & if \quad m = n+1 \\ 0 & otherwise \end{cases} \qquad [x_{mn}] = \begin{cases} x & if \quad m = n-1 \\ y & if \quad m = n \\ z & if \quad m = n+1 \\ 0 & otherwise \end{cases}$$

, , $(f)_{j+1}$ and $(f)_j$ are known and unknown vectors respectively. It is a tridiagonal system and for its efficient numerical computation, well known Thomas algorithm

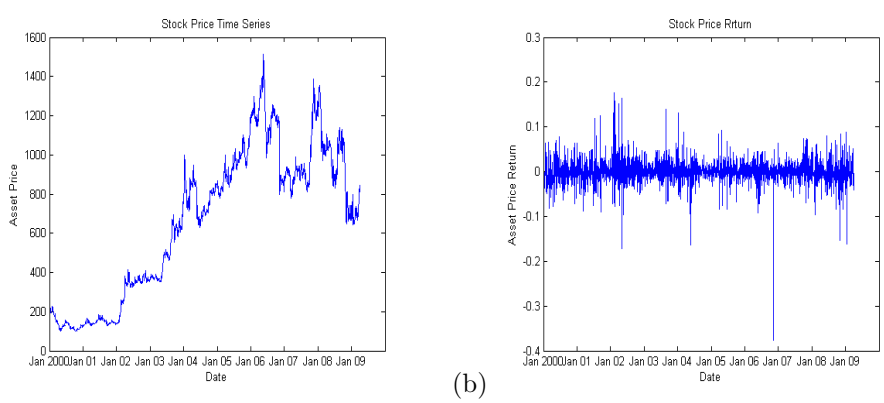
is suggested by Kwok([12]). The algorithm is an efficient implementation of the Gaussian elimination procedure and is numerically stable if $a_j < 0$, $c_j < 0$ and $b_j > |a_j| + |c_j|$. Fortunately, these conditions are automatically fulfilled by the tridiagonal system Eq.(1.3.1).

1.4. **Sensitivities by Finite Difference Scheme.** The finite difference method provides the price of an option and gives the option price at each $(M+1)\times(N+1)$ grids of the domain, which have a mathematical relevance and importance itself and financial aspects also. The $(N+1)$th column is sufficient to determine the option price, but this does not mean that the remaining values are useless, also significantly important in the determination the sensitivity i.e Greeks, directly on grid without having to go through the lengthy procedure. Since a finite difference scheme provide approximation for the derivatives with respect to S and t , therefore we can calculate the delta , gamma and theta using values associated with grids as:

$$Delta(\triangle) = \frac{V^{i+1}-V^{i}}{\triangle S},\ Gamma(\gamma) = \frac{V^{i+1}-2V^{i}+V^{i-1}}{\triangle S^2},\ Theta(\theta) = \frac{V^{j+1}-V^{j}}{\triangle t}$$

2. Results and Discussion

The parameters of EGARCH(1,1) model are estimated for Oil and Natural Gas corporation (ONGC) Ltd. using the 2311 data points of stock price presented through figure(1a) and whereas, figure(1b) represents the percentage change of stock return. The estimated parameters are as $\varepsilon_t = 0.00050494$, $\alpha_0 = -0.49553$, $\alpha_1 = 0.26875$, $\beta_1 = 0.93225$ and $\gamma_1 = 0.091203$ and volatility is forecasted using these estimated parameters from Eqs.(1.0.4-1.0.5), which is represented through table(1). Using forecasted volatility the option prices and sensitivity parameters for different stock price S, strike prices $K = \$20, \$22, \$24, \$26$, risk-free $r = 8\%, 10\%, 12\%, 14\%$ interest rate and dividends $D = 10\%, 15\%, 20\%, 25\%$ at four different maturity ($T = 1, 2, 3, 4$ months), are determined and results are shown graphically and by tabular forms.



(a) (b)

Figure 1. (a) Stock Price (b) Stock Price Return of ONGC from 03/01/2000 to 30/03/2009

Figure(2 a) shows that the put premium increases as strike price increases and table(3) shows that option prices are proportional to strike prices, a 10% increment in strike price is a cause to goes up 10% option value. Inspite of increasing nature

No. Days	Volatility (σ_i)	No. Days	Volatility (σ_i)	No. Days	Volatility (σ_i)	No. Days	Volatility (σ_i)
1	0.4829	16	0.4283	31	0.4101	46	0.4046
2	0.4769	17	0.4264	32	0.4095	47	0.4044
3	0.4714	18	0.4247	33	0.4090	48	0.4042
4	0.4663	19	0.4231	34	0.4080	49	0.4040
5	0.4616	20	0.4202	35	0.4080	50	0.4038
6	0.4572	21	0.4189	36	0.4076	51	0.4037
7	0.4532	22	0.4177	37	0.4072	52	0.4035
8	0.4495	23	0.4166	38	0.4068	53	0.4034
9	0.4461	24	0.4156	39	0.4064	54	0.4033
10	0.4430	25	0.4146	40	0.4061	55	0.4032
11	0.4400	26	0.4137	41	0.4058	56	0.4031
12	0.4373	27	0.4129	42	0.4055	57	0.4030
13	0.4348	28	0.4121	43	0.4052	58	0.4029
14	0.4325	29	0.4114	44	0.4050	59	0.4028
15	0.4303	30	0.4107	45	0.4048	60	0.4027

TABLE 1. 60 Days Forecasted Volatility of Reliance Ltd. using parameters $\varepsilon_t = 0.00050494$, $\alpha_0 = -0.49553$, $\alpha_1 = 0.26875$, $\beta_1 = 0.93225$ and $\gamma_1 = 0.091203$ of EGARCH(1,1) Model

	(S)	($)	=	0.0000	18.0000	20.0000	22.0000	24.000	26.0000	30.0000
T in Days	D in (%)	r in (%)	K in ($)				P			
			20	20.0000	2.2588	0.9640	0.3052	0.0726	0.0134	0.0003
30	10	8	22	22.0000	4.0795	2.3271	1.0605	0.3774	0.1058	0.0045
			24	24.0000	6.0461	4.1081	2.3996	1.1570	0.4530	0.0386
			26	26.0000	8.0440	6.0558	4.1428	2.4755	1.2535	0.1894
		10		20.0000	2.2311	0.9464	0.2975	0.0702	0.0129	0.0002
30	10	12	20	20.0000	2.2036	0.9291	0.2900	0.0679	0.0124	0.0002
		14		20.0000	2.1763	0.9120	0.2826	0.0657	0.0119	0.0002
	15			20.0000	2.3287	1.0089	0.3251	0.0788	0.0149	0.0003
30	20	10	20	20.0000	2.3997	1.0551	0.3461	0.0855	0.0164	0.0003
	25			20.0000	2.4716	1.1026	0.3681	0.0926	0.0182	0.0004
60				20.0000	2.5423	1.3460	0.6209	0.2526	0.0923	0.0096
90	10	8	20	20.0000	2.7973	1.6544	0.9006	0.4557	0.2168	0.0425
120				20.0000	3.0268	1.9195	1.1508	0.6572	0.3604	0.0988

TABLE 2. Option Price for different Stock Price S, Strike Price $K = \$20, \$22, \$24, \$26$, Interest rates $r = 8\%, 10\%, 12\%, 14\%$, Dividends $D = 10\%, 15\%, 20\%, 25\%$ and Maturity $T = 1, 2, 3$ and 4 months

of option premium, option position is more profitable as compared to stock position in case of profit and loss both.

The interest rate (r) has an inverse effect on put premium as r increases the value of put premium decreases. From the table(4) it is observed that the increment of 2% interest rate is a cause to increase the profit and to reduce the loss in option position.

Figure(2 c) and table(5) depict the impact of dividends on option pricing. The put premium increases as dividend increases. In this case the investment ratio, profit ratio and loss ratio decrease as dividend increases. The effect of time of maturities is represented through figure(2 d) and table(6), and investment ratio, profit ratio and loss ratio decrease as time of maturities increase.

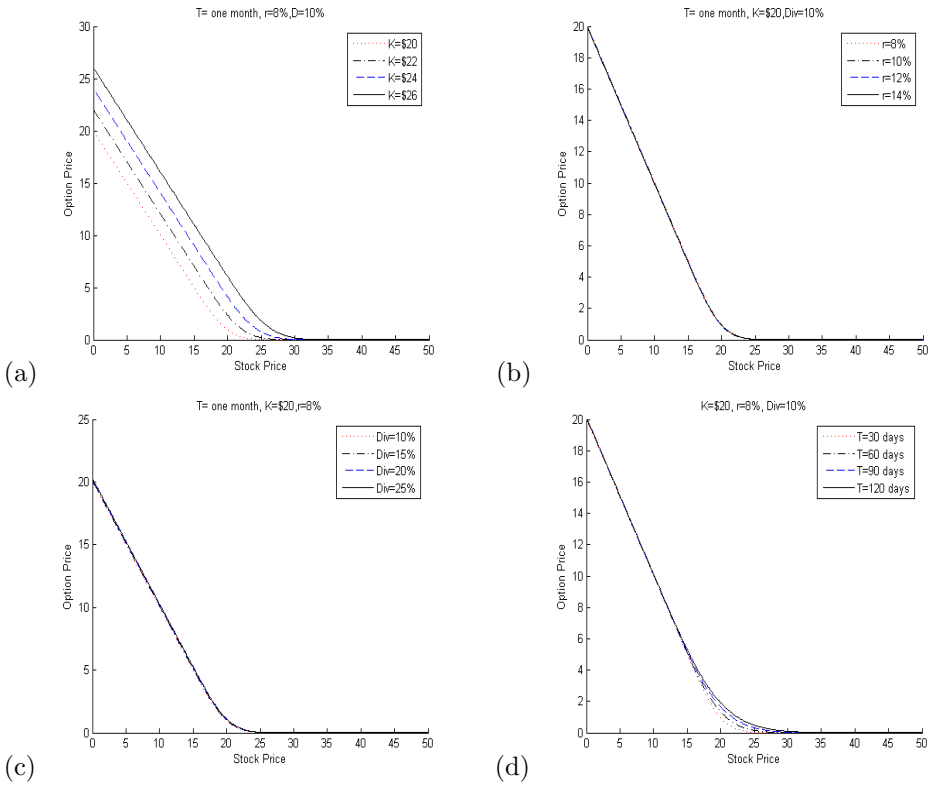


FIGURE 2. Option Price for (a) K=\$20,\$22,\$24,\$26, r=8%, Div=10% and T=1 month (b) K=\$20, r=8%,10%,12%,14%, Div=10% and T=1 month (c) K=\$20, r=8%, Div=10%,15%,20%,25% and T=1 month (d) K=\$20, r=8%, Div=10% and T=1,2,3,4 months for different Stock Prices S

Since option pricing depicts a strange paradox in different situations, hence, this is a confusing stage for a investor/institution. Here G̈reeks" will be helpful to rescue from such situations.

Delta plays a crucial role in the hedging of portfolio. The values of delta of put premium represented through figures(3 a-3 d) for different parameters. The $\triangle$ curves are like a step function equal to 0 above the strike price and -1 below it. When $\triangle$ curve is a horizontal line then hedging is static, which remain in a larger and larger interval. When $\triangle$=-1, then the portfolio can be hedged by holding the underlying equal to option and when $\triangle$=0, then portfolio can be hedged without holding any number of asset. The delta going from almost -1 to almost 0 in a shorter and shorter interval, which lead to dynamic hedging. The less inclination of the delta curve represent less dynamic hedging character. But as the time of maturities decrease the hedging become more and more dynamic. The negativity of delta curves means that a long position in a put option should be hedged by a continually varying long position in the multiple of $\triangle$ of underlying asset. Hence once can make money using delta hedging as hold a long option position with holding of $\triangle$ units of asset of underlying.

Figures(4 a-4 d) represent the value of theta (θ) of put option. Theta is usually negative for an option with exception of in-the-money European put option. This

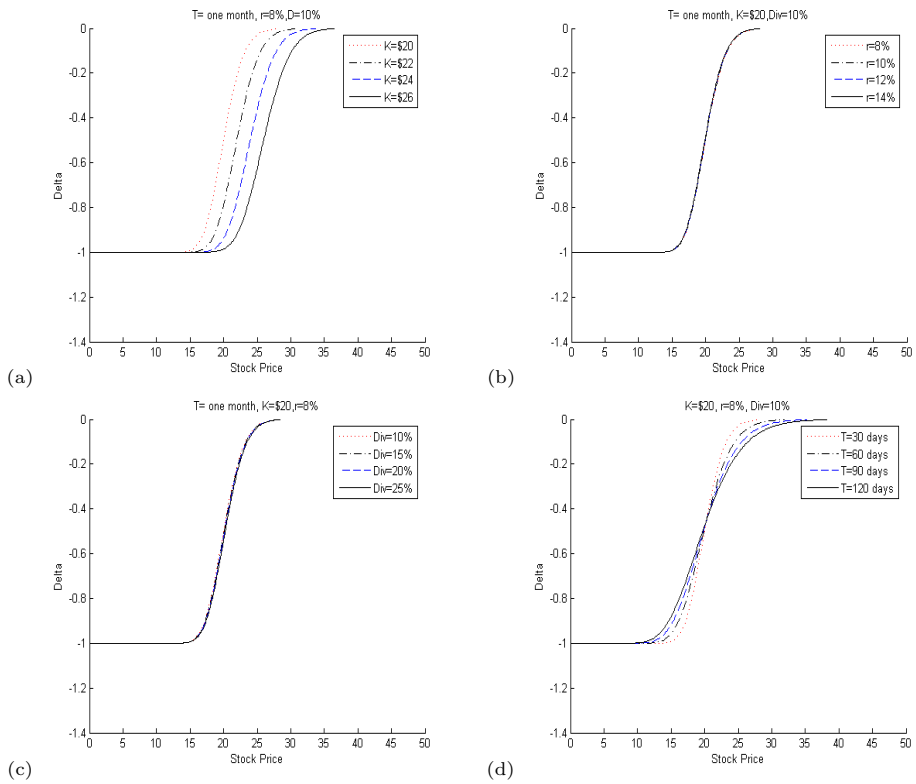


FIGURE 3. Delta ($\triangle_P$) of the Put Option for (a) K=\$20,\$22,\$24,\$26, r=8%, Div=10% and T=1 month (b) K=\$20, r=8%,10%,12%,14%, Div=10% and T=1 month (c) K=\$20, r=8%, Div=10%,15%,20%,25% and T=1 month (d) K=\$20, r=8%, Div=10% and T=1,2,3,4 months for different Stock Prices S

is because as time to maturity decreases with all else remaining same, the option tends to become less valuable. The theta has its greatest absolute value when the put option is at-the-money. When put option is in-the-money, theta is close to zero. From the figures (4 a-4 d), it is clear that put option become less valuable as strike price, interest rate, dividends and maturity increase.

Variation of gamma (γ) is presented through figs.(5 a-5 d) for various parameters K, r, D and T. The curve of γ for each parameters are almost symmetric about $S = K$ (at-the-money), above the S-axis. The impact of dividend and rate of interest on gamma is insignificant but it is more influenced by K and T. Away from the strike price (deep in-the-money or deep out-of-the-money) the γ becomes zero this means that delta hedging remains static. The peakness of γ decreases as K and T increase which show that option position is more sensitive at-the-money and hence $\triangle$ hedging is more dynamic at this movement. Therefore financial analyst should be more conscious at such situation and risk arises due to change of stock price can be controlled by applying $\triangle$ hedging. On the other hand lack of awareness may lead to drastic/undesired changes in portfolio.

	Stock Price (S)($)	20.00	22.00	24.00	26.00
	Strike Price (K)($)	20.00	22.00	24.00	26.00
	Option Price	0.9640	1.0605	1.1570	1.2535
Stock Position	Investment	2000.00	2200.00	2400.00	2600.00
	Interest on Investment	13.3333	14.6667	16.0000	17.3333
	Total Investment	2013.33	2214.67	2416.00	2617.33
Option Position	Investment	96.4000	106.0500	115.7000	125.3500
	Interest on Investment	0.6426	0.7070	0.7713	0.8356
	Total Investment	97.0427	106.757	116.4713	126.1857
Investment Ratio	(Stock V/S Option)	20.7468	20.7449	20.7433	20.7419
	(%) Ratio (Stock V/S Option)	4.8521	4.8526	4.8530	4.8533
Profit	if Stock (S)($)	18.00	20.00	22.00	24.00
Stock Position		213.3333	214.6667	216.0000	217.3333
	(%) Profit	10.6667	9.7575	9.0000	8.3589
Option Position		128.8373	125.413	123.4887	121.3643
	(%) Profit	132.76360	117.4752	106.0249	96.1791
Profit Ratio	(Stock V/S Option)	1.6558	1.7117	1.7491	1.7908
Loss	if Stock (S)($)	22.00	24.00	26.00	28.00
	Stock Position	-186.6667	-185.3333	-184.0000	-182.6667
	Option Position	-97.0427	-106.7570	-116.4713	-126.1857
Loss Ratio	(Stock V/S Option)	1.9236	1.7360	1.5798	1.4476

TABLE 3. Table for option and stock position for different strike price, No. of Share=100, Interest rate $r = 8\%$, Maturity $T = 1$month and Dividend $D = 10\%$

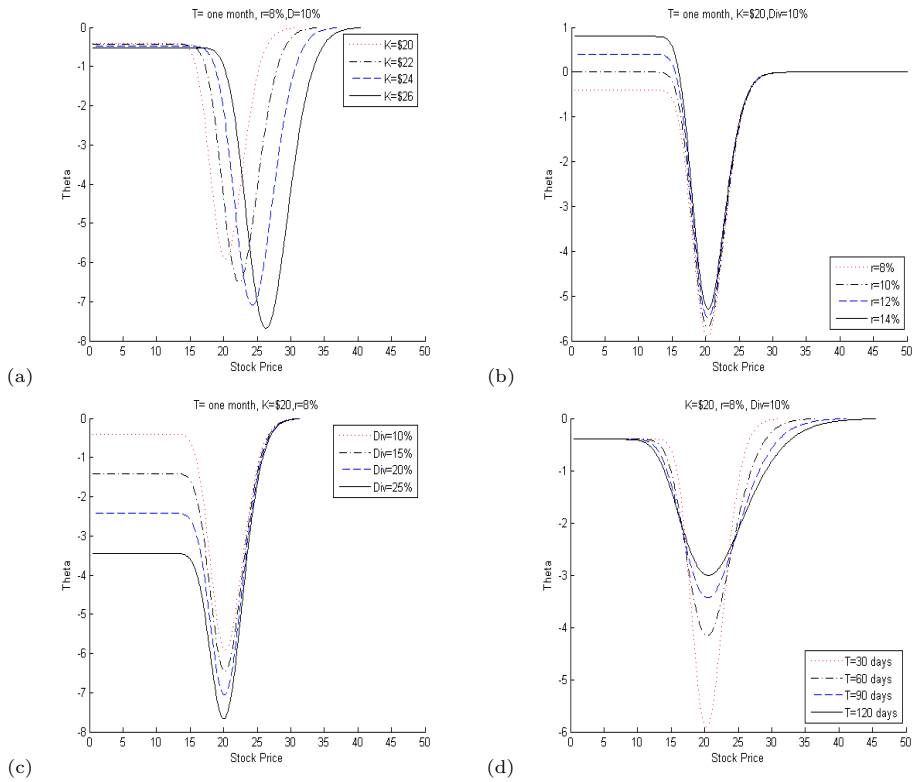


FIGURE 4. Theta (θ_P) of the Put Option for (a) K=\$20,\$22,\$24,\$26, r=8%, Div=10% and T=1 month (b) K=\$20, r=8%,10%,12%,14%, Div=10% and T=1 month (c) K=\$20, r=8%, Div=10%,15%,20%,25% and T=1 month (d) K=\$20, r=8%, Div=10% and T=1,2,3,4 months for different Stock Prices S

	Interest Rate (r)(%)	8.00	10.00	12.00	24.00
	Option Price	0.9640	0.9464	0.9291	0.9120
Stock Position	Investment	2000.00	2000.00	2000.00	2000.00
	Interest on Investment	13.3333	16.6667	20.0000	23.3333
	Total Investment	2013.3333	2016.6667	2020.0000	2023.3333
Option Position	Investment	96.4000	94.6400	92.9100	91.2000
	Interest on Investment	0.6427	0.7887	0.9291	1.0640
	Total Investment	97.0427	95.4287	93.8391	92.2640
Investment Ratio	(Stock V/S Option)	20.7469	21.1327	21.5262	21.9298
	(%) Ratio (Stock V/S Option)	4.8521	4.7714	4.6920	4.6132
Profit	if Stock (S)($)	18.00			
Stock Position		213.3333	216.6667	220.0000	223.3333
	(%) Profit	10.6667	10.8333	11.0000	11.1667
Option Position		128.8373	127.6813	126.5209	125.3660
	(%) Profit	132.7636	133.7977	134.8275	135.8775
Profit Ratio	(Stock V/S Option)	1.6558	1.6969	1.7388	1.7815
Loss	if Stock (S)($)		22.00		
	Stock Position	-186.6667	-183.3333	-180.0000	-176.6667
	Option Position	-97.0427	-95.4287	-93.8391	-92.2640
Loss Ratio	(Stock V/S Option)	1.9236	1.9212	1.9182	1.9148

TABLE 4. Table for option and stock position for different interest rate, No. of Share=100,Strike Price K = \$20, Maturity T = 1month and Dividend D = 10%

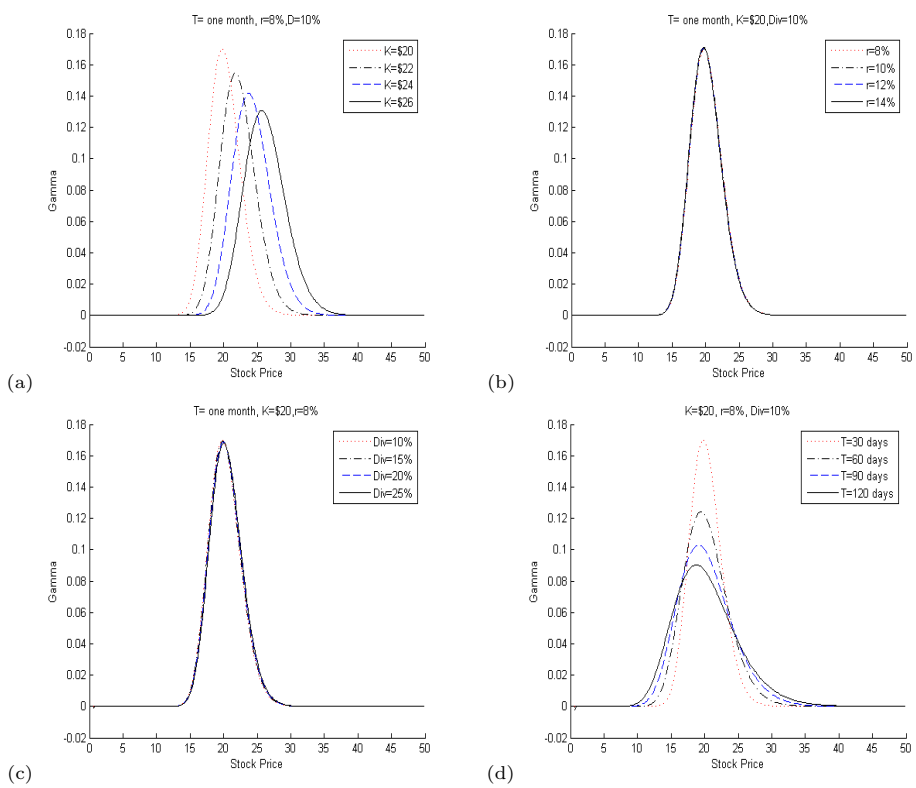


FIGURE 5. Gamma (γ_P) of the Put Option for (a) K=\$20,\$22,\$24,\$26, r=8%, Div=10% and T=1 month (b) K=\$20, r=8%,10%,12%,14%, Div=10% and T=1 month (c) K=\$20, r=8%, Div=10%,15%,20%,25% and T=1 month (d) K=\$20, r=8%, Div=10% and T=1,2,3,4 months for different Stock Prices S

3. CONCLUSION

The variations in interest rates and dividends have no significant impact on put option price, while the increase in interest rates and dividends, is cause of high

	Dividends (D)(%)	10.00	15.00	20.00	25.00
	Option Price	0.9640	1.0089	1.0551	1.1026
Stock Position	Investment	2000.00	2000.00	2000.00	2000.00
	Interest on Investment	13.3333	13.3333	13.3333	13.3333
	Total Investment	2013.3333	2013.3333	2013.3333	2013.3333
Option Position	Investment	96.4000	100.8900	105.5100	110.2600
	Interest on Investment	0.6427	0.6726	0.7034	0.7351
	Total Investment	97.0427	101.5626	106.2134	110.9951
Investment Ratio	(Stock V/S Option)	20.7469	19.8236	18.9555	18.1389
	(%) Ratio (Stock V/S Option)	4.8521	5.0781	5.3107	5.5498
Profit	if Stock (S)($)	18.00			
Stock Position		213.3333	213.3333	213.3333	213.3333
	(%) Profit	10.6667	10.6667	10.6667	10.6667
Option Position		128.8373	131.3074	133.7566	136.1649
	(%) Profit	132.7636	129.2872	125.9319	122.6766
Profit Ratio	(Stock V/S Option)	1.6558	1.6247	1.5949	1.5667
Loss	if Stock (S)($)		22.00		
	Stock Position	-186.6667	-186.6667	-186.6667	-186.6667
	Option Position	-97.0427	-101.5626	-106.2134	-110.9951
Loss Ratio	(Stock V/S Option)	1.9236	1.8379	1.7575	1.6818

TABLE 5. Table for option and stock position for different Dividends D, No. of Share=100, Strike Price $K = \$20$, Maturity $T = 1month$ and Interest rate $r = 10\%$

	Maturity Time (T)(in months)	1	2	3	4
	Option Price	0.9640	1.0089	1.0551	1.1026
Stock Position	Investment	2000.00	2000.00	2000.00	2000.00
	Interest on Investment	13.3333	26.6667	40.0000	53.3333
	Total Investment	2013.3333	2026.6667	2040.0000	2053.3333
Option Position	Investment	96.4000	100.8900	105.5100	110.2600
	Interest on Investment	0.6427	0.6726	0.7034	0.7351
	Total Investment	97.0427	101.5626	106.2134	110.9951
Investment Ratio	(Stock V/S Option)	20.7469	19.9549	19.2066	18.4993
	(%) Ratio (Stock V/S Option)	4.8521	5.0781	5.3107	5.5498
Profit	if Stock (S)($)	18.00			
Stock Position		213.3333	226.6667	240.0000	253.3333
	(%) Profit	10.6667	11.3333	12.0000	12.6667
Option Position		128.8373	153.6674	173.5166	191.6849
	(%) Profit	132.7636	150.3185	163.3660	172.6968
Profit Ratio	(Stock V/S Option)	1.6558	1.4847	1.3832	1.3216
Loss	if Stock (S)($)		22.00		
	Stock Position	-186.6667	-173.3333	-160.0000	-146.6667
	Option Position	-97.0427	-101.5626	-106.2134	-110.9951
Loss Ratio	(Stock V/S Option)	1.9236	1.7067	1.5064	1.3214

TABLE 6. Table for option and stock position for different maturity period T, No. of Share=100, Strike Price $K = \$20$, Interest rate $r = 8\%$ and Dividend $D = 10\%$

volatility. The increment in interest rate is profitable for put option position as well as short sell of stock position. The results of this study suggests that put option position is less risky with compared to stock position and a little amount of money can make a position in option as compared to stock position.

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